Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (Bsc.IT Semester III)

Data Structures Practical

Practical - IX

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Class : SYIT	Batch : 1
Date of Assignment : 19/08/2026	Date/Time of Submission : 19/08/2026

PART B - QUESTIONS

Searching Algorithms:

Write programs to implement and compare:

a. binary search

code:

```
def binarySearch(array, x, low, high):

    # Repeat until the pointers low and high meet each other
    while low <= high:

        mid = low + (high - low)//2

        if x == array[mid]:
            return mid

        elif x > array[mid]:
            low = mid + 1

        else:
            high = mid - 1

    return -1


array = [3, 4, 5, 6, 7, 8, 9]
x = 4

result = binarySearch(array, x, 0, len(array)-1)

if result != -1:
    print("Element is present at index " + str(result))
else:
    print("Not found")
```

output:

```
>>
===== RESTART: C:/Users/info/Downloa
Element is present at index 1
>>
```

b.linear search (on a sorted array)

code:

```
def linearSearch(array, n, x):

    # Going through array sequencially
    for i in range(0, n):
        if (array[i] == x):
            return i
    return -1


array = [2, 4, 0, 1, 9]
x = 1
n = len(array)
result = linearSearch(array, n, x)
if(result == -1):
    print("Element not found")
else:
    print("Element found at index: ", result)
```

output:

```
===== RESTART: C:/Users/info/Dc
Element found at index:  3
>>
```

Curiosity Questions:

1. Why is Binary Search faster than Linear Search for a large sorted array?
2. What will happen if we apply Binary Search to an unsorted array?
3. Can Linear Search be used when the array contains duplicate elements? What will it return?
4. If an array has 1,000 elements, how many elements might Linear Search check in the worst case?
5. Why does Binary Search always check the middle element?
6. What happens when the element being searched is smaller than the middle element in Binary Search?

7. Can Binary Search be used to search a single element array? Why?
8. Which search would you prefer for a small unsorted list, and why?
9. If the same sorted array is searched many times, why might Binary Search be a better choice?
10. If Linear Search and Binary Search are given the same array and same element, will they always take the same number of comparisons? Why?